Class 5

Science

Solutions

Olympiad Program

Table of Content

1.	Matter Around Us	1 – 7
2.	Rocks and Minerals	8 – 11
3.	Food and Health	12 – 15
4.	Diseases	16 – 18
5.	Plants Around Us	19 – 22
6.	Animals Around us	23 – 26
7.	Human Body	27 – 30
8.	Work and Energy	31 – 35
9.	Force and Simple Machines	36 – 39
10.	Sun, Moon and Earth	40 – 44
11.	Air and Water	45 – 49
12.	Pollution and Natural Disasters	50 – 53

1. Matter around us

Worksheet – 1

1. Volume of water left
Initial water = 500 mL. Water poured out = 200 mL.
Remaining = 500 − 200 = 300 mL
Answer: C) 300 mL

2. Material properties (shiny, conducts electricity, drawn into wires)
Metals are lustrous (shine), conduct electricity, and are ductile (can be drawn into wires).
Answer: C) Metal

3. Does not conduct electricity and does not allow light to pass
Wood is an insulator (does not conduct electricity) and is opaque (does not allow light to pass).
Answer: C) Wood

4. Material that allows light to pass clearly
Transparent materials allow light to pass through clearly (e.g., glass, clean water).
Answer: B) Transparent

5. Correct property matching
Density is a physical property.
Elasticity → mechanical property
Conductivity → physical/electrical property
Transparency → optical property
Answer: B) Density – Physical property

6. Salt dissolving in water
Solubility is the ability of a substance (salt) to dissolve in a liquid (water).
Answer: B) Solubility

7. Amount of rice used
Initial mass = 5 kg = 5000 g
Remaining = 3200 g
Used = 5000 − 3200 = 1800 g = 1.8 kg
Answer: B) 1.8 kg

8. Glass breaks easily, steel does not
Brittleness is the property of a material to break easily when force is applied.
Answer: B) Brittleness

9. Copper used in wires
Ductility is the ability to be drawn into thin wires (like copper wires).
Answer: B) Ductility

10. Material for helmet (hard + strong)
Diamond is extremely hard (resists scratches), and steel is strong (withstands heavy impact).
Answer: B) Diamond (Hard) and Steel (Strong)

Worksheet-2

1. Definition of matter
Matter is anything that has mass and takes up space (volume).
Answer: B) Anything that occupies space and has mass

2. Not a state of matter
Solid, liquid, and gas are states of matter. Light is energy, not matter.
Answer: D) Light

3. Particles move most freely
Gas particles have the highest kinetic energy and move freely in all directions.
Answer: C) Gas

4. Easily compressed state
Gas particles are far apart, so they can be compressed easily.
Answer: C) Gas

5. "Shape-shifter" state
Liquids do not have a fixed shape and take the shape of their container.
Answer: B) Liquid

6. Solid to liquid process
Melting is the change of state from solid to liquid on heating.
Answer: C) Melting

7. Fixed shape and fixed volume
Solids have tightly packed particles, so they maintain both shape and volume.
Answer: D) Solid

8. Solid changes directly into gas
Sublimation is when a solid turns directly into gas without becoming liquid. Dry ice (solid CO_2) shows this property.
Answer: B) Sublimation; Dry Ice

9. Meaning of sublimation
Sublimation skips the liquid stage completely.
Answer: C) A solid changes directly into a gas

10. Boiling at high altitudes
At higher altitudes, atmospheric pressure is low, so liquids boil at lower temperatures.
Answer: A) The air pressure is lower, so less heat is needed for boiling

---

Worksheet – 3

1. Water in which sugar is dissolved
The substance that dissolves the solute (sugar) is called the solvent (water).
Answer: B) Solvent

2. Mixture with visible parts
In heterogeneous mixtures, components are clearly visible (e.g., sand + water).
Answer: C) Heterogeneous mixture

3. Sand and water mixture
Sand is insoluble in water, so it settles down after mixing.
Answer: C) Sand settles at the bottom

4. Example of a mixture
Salt water is a mixture of salt (solute) and water (solvent).
Answer: A) Salt water

5. Homogeneous mixture
Air is a uniform mixture of gases, so it is homogeneous.
Answer: D) Air

6. Homogeneous mixture property
Homogeneous mixtures are uniform in composition everywhere.

Answer: A) The composition and properties are uniform throughout the mixture.

7. Tincture of iodine
Iodine dissolves in alcohol, so iodine is solute and alcohol is solvent.
Answer: B) Iodine is the solute and alcohol is the solvent

8. Water, oil, and alcohol layers
Alcohol is miscible with water, but oil is immiscible, so layers form.
Answer: C) Alcohol mixes with water but oil does not mix with them

9. Milk and water mixture
Liquids that mix completely and form a uniform mixture are miscible.
Answer: B) Miscible

10. Liquid X behavior
Since it mixes with water but forms layers with oil, it is miscible with water only.
Answer: C) Liquid X is miscible with water but immiscible with oil

---

Worksheet – 4

1. Separating sand from water
Filtration is used to separate an insoluble solid (sand) from a liquid (water) using a filter.
Answer: A) Filtration

2. Iron filings and sulphur powder
Iron is magnetic, but sulphur is not. A magnet can easily separate iron filings.
Answer: B) Magnetic Separation

3. Heating a liquid, vaporising and then cooling
Explanation: Distillation involves evaporation + condensation to separate and collect pure liquid.
Answer: C) Distillation

4. Sand and small rocks
Sieving separates particles based on size using a mesh.
Answer: B) Sieving

5. Oil and water
Oil and water are immiscible, so they form layers and can be separated by settling/decantation.
Answer: D) Letting them settle

6. Rice and salt
Rice grains are large and visible, so they can be separated manually from salt.
Answer: C) Handpicking

7. Least effective method for sand and salt
Neither sand nor salt is magnetic, so this method does not work.
Answer: B) Magnetic Separation

8. Winnowing process
Lighter chaff is blown away by wind, while heavier grains fall due to higher density.
Answer: B) Because of its density

9. Sugar and salt in water
First, filtration removes insoluble impurities (if any), then evaporation separates dissolved solids.
Answer: A) Filtration followed by Evaporation

10. Odd one out
Distillation, separating funnel, and evaporation are lab-based separation methods, while winnowing is a traditional field method.
Answer: C) Winnowing

Competitive Corner

1. Separation of cream from milk
Centrifugation spins the mixture at high speed, separating lighter cream from heavier milk.
Answer: A) Centrifugation

2. Substance that undergoes sublimation
Naphthalene directly changes from solid to gas without becoming liquid.
Answer: B) Naphthalene

3. Physical change
It changes only shape/size, not composition, so it is a physical change.
Answer: C) Tearing of paper

4. Best conditions for gas → liquid
High pressure brings particles closer and low temperature reduces kinetic energy, helping condensation.
Answer: A) High pressure, low temperature

5. Property allowing gases to be compressed
Gas particles are far apart, so they can be compressed easily.
Answer: A) Large intermolecular spaces

6. Example of condensation
Water vapour in air cools and changes into liquid droplets.
Answer: C) Water droplets forming on the outside of a cold glass

7. Gas changing directly into solid
Deposition is gas → solid without becoming liquid first.
Answer: B) Deposition

8. Overlap of homogeneous mixtures and miscible solutions
Both form uniform mixtures where components cannot be seen separately.
Answer: A) Components mix completely to form a single phase

9. Separation of acetone and water
Distillation separates liquids based on different boiling points.
Answer: C) Distillation

10. Cashew nuts and flour separation
Mesh allows small flour particles to pass through while retaining larger cashew pieces.
Answer: C) Difference in their particle sizes

11. Flow chart processes
Solid → Liquid = Melting
Liquid → Gas = Evaporation
Gas → Liquid = Condensation
Answer: B) P-Melting, Q-Evaporation, R-Condensation

12. Iron, glass, wood rods in furnace
Iron conducts heat very well, so heat travels quickly through it.
Answer: B) The Iron rod will be the hottest at the far end due to high thermal conductivity.

Multiple Correct answer:

13. Change of state from solid
Correct concept key:
Solid → Liquid = Melting
Solid → Gas = Sublimation
Both melting and sublimation involve a solid changing its state, so they are correct processes involving solids.
Answer: A) Melting and B) Sublimation (both involve solid change)

14. True statements about solutions
A Solution is a homogeneous mixture
B Solute dissolves in solvent
D Salt solution is a valid example
Answer: A, B and D are correct (C is incorrect)

15. Heterogeneous mixtures
(A) Oil and water
(B) Sand and water
(D) Kerosene and water
Heterogeneous mixtures have visible separate components or layers.
Answer: A, B and D

16. Separation techniques used in daily life
All given methods are commonly used:
Handpicking → removing impurities manually
Winnowing → separating lighter and heavier particles
Sieving → separating based on size
Distillation → used in purification (e.g., water)
Answer: A, B, C, D (All correct)

Assertion and reason type:

17. Answer: A) Both assertion and reason are true, and the reason is the correct explanation for the assertion.
Explanation:

- Assertion is true: Condensation is the process where a gas changes into a liquid.
- Reason is also true: During condensation, gas particles lose energy, slow down, and come closer together, forming a liquid.
- The reason correctly explains why condensation happens.

18. Answer: D) The assertion is false, but the reason is true.
Explanation:

- Assertion is false: When ice melts, it changes from **solid to liquid**, not gas.
- Reason is true: Melting occurs when a substance **absorbs heat energy**.
- So, only the reason is correct, but it does not support the incorrect assertion.

Match the following

19. Answer: P–b, Q–a, R–c, S–d
Explanation:

- Solute is the substance dissolved (e.g., sugar)
- Solvent dissolves the solute (e.g., water)
- Solution is a uniform mixture
- Miscible liquids mix completely (e.g., alcohol + water)
-

20. Answer: P–c, Q–d, R–a, S–b

Explanation:

P. Melting of ice → c (Change in state without new substance)

Q. Rusting of iron → d (Change in color with new substance formed)

R. Burning of paper → a (Production of heat and light)

S. Breaking a glass → b (Change in shape or size)

Comprehension:

21. Property showing how easily a material can be stretched

Answer: B) Elasticity

Explanation: Elasticity is the ability of a material to be stretched and return to its original shape.

22. Copper can be drawn into thin wires

Answer: B) Ductility

Explanation: Ductility is the property of metals to be drawn into thin wires (e.g., copper wires).

23. Glass breaks easily under force

Answer: D) Brittleness

Explanation: Brittleness is the tendency of a material to break easily without much deformation.

24. State with definite shape and volume

Answer: C) Solid

Explanation: Solids have tightly packed particles, so they maintain both shape and volume.

25. State with definite volume but no fixed shape

Answer: B) Liquid

Explanation: Liquids take the shape of their container but keep a fixed volume.

26. Particles move freely and fill container

Answer: C) Gas

Explanation: Gas particles move freely in all directions and occupy the entire space available.

Previous Year Questions

1. Correct Answer: D. Water and refined oil

Explanation:

Immiscible liquids are those that do not mix with each other and form separate layers. Water and refined oil do not mix and oil floats on water due to lower density. Mango juice, milk, and vinegar are all water-based liquids, so they are miscible with water.

2. Wooden spoon – incorrect statement

Correct Answer: C. It can be compressed

Explanation:

Wood is a solid material and has:

- poor compressibility (cannot be easily compressed)

3. Physical and chemical changes

Correct Answer: D. Formation of ice (Physical) and Burning of wood (Chemical)

Explanation:

- Formation of ice → physical change (only state changes, no new substance formed)

- Burning of wood → chemical change (new substances like ash and gases are formed) Other options mix up physical and chemical changes incorrectly.

4. Separation methods

Correct Answer: B. Soil can be separated by sedimentation and decantation

Explanation:

- Soil → settles down in water, so sedimentation and decantation is correct

5. Water bottle frozen to rim

Correct Answer: C. Freezing increases volume of water

Explanation:

When water freezes, it expands because ice has a more open structure than liquid water. This increases the volume, which is why a completely filled bottle may crack or bulge.

---

2. Rocks and minerals

Worksheet-1

1. **Answer: B) Crust**
   The crust is the outermost solid layer of the Earth where all landforms, oceans, and life exist. It is the thinnest layer compared to others.
2. **Answer: C) Mantle**
   The mantle lies below the crust and is the thickest layer of the Earth. It extends to about 2900 km and is made of semi-solid rock.
3. **Answer: C) Mantle**
   Convection currents in the mantle cause the movement of tectonic plates on the crust.
4. **Answer: B) Magma from the mantle coming to the surface**
   When pressure builds up, magma from the mantle forces its way through cracks in the crust, causing eruptions.
5. **Answer: C) Outer Core**
   The outer core is made of molten iron and nickel, whose movement generates Earth's magnetic field.
6. **Answer: B) It is under high pressure**
   The extreme pressure at the center of Earth keeps the inner core solid despite very high temperatures.
7. **Answer: C) Granite**
   Granite forms when magma cools and solidifies slowly beneath the Earth's surface.
8. **Answer: B) Marble**
   Limestone changes into marble due to heat and pressure, making it a metamorphic rock.
9. **Answer: A) Rock Cycle**
   The rock cycle describes how rocks are formed, broken down, and transformed from one type to another.
10. **Answer: B) Metamorphism**
    Metamorphism occurs when rocks are changed by heat and pressure without melting.

Worksheet – 2

1. **Answer: B) Naturally occurring substances with a definite chemical composition**
   Minerals are naturally formed substances found in the Earth's crust. They have a fixed chemical composition and structure, like iron, gold, and salt.
2. **Answer: C) Iron**
   Iron is a metallic mineral because it contains metals and is used in industries for making tools, machines, and construction materials.
3. **Answer: B) Gold**
   Gold is a precious metal used in jewellery due to its shine, malleability, and resistance to corrosion.
4. **Answer: C) Nitrogen**
   Nitrogen makes up about 78% of the Earth's atmosphere, making it the most abundant gas.
5. **Answer: B) Oxygen**
   Oxygen is required by living organisms for respiration, which produces energy.

6. **Answer: D) Atmosphere**
   The atmosphere is the layer of gases surrounding the Earth, containing air necessary for life.
7. **Answer: C) 71%**
   About 71% of Earth's surface is covered by water, mostly in oceans.
8. **Answer: C) Well**
   A well is dug by humans to access underground water, making it a man-made source.
9. **Answer: C) Loamy soil**
   Loamy soil is a mixture of sand, clay, and humus, making it fertile and ideal for agriculture.
10. **Answer: C) Minerals, organic matter, water, and air**
    Soil is a mixture of rock particles, decayed organic material, water, and air, which together support plant growth.

Competitive Corner

1. **Answer: D) Inner Core**
   The inner core is made mainly of iron and nickel and remains solid due to extreme pressure.
2. **Answer: B) Mantle**
   The mantle is the thickest layer, extending about 2900 km beneath the crust.
3. **Answer: B) Igneous**
   Igneous rocks form when magma or lava cools and solidifies.
4. **Answer: C) Erosion and deposition**
   Sedimentary rocks form when sediments are deposited, compacted, and cemented over time.
5. **Answer: C) Marble**
   Marble forms when limestone is subjected to heat and pressure.
6. **Answer: C) Nitrogen**
   Nitrogen makes up about 78% of the atmosphere.
7. **Answer: D) Photosynthesis**
   Photosynthesis is related to plants, not directly part of the water cycle (which includes evaporation, condensation, and precipitation).
8. **Answer: A) Limestone**
   Limestone is widely used in building materials and sculptures.
9. **Answer: B) Groundwater**
   Groundwater is stored beneath the Earth's surface in soil and rock spaces.
10. **Answer: B) Graphite**
    Graphite is a form of carbon used as the "lead" in pencils.
11. **Answer: B) Afforestation (planting trees)**
    Planting trees helps hold soil together and prevents erosion.
12. **Answer: C) Coniferous forests**
    Coniferous forests have needle-like leaves and are adapted to cold climates.
13. **Answer: B) Planting more trees (afforestation)**
    Afforestation helps restore forest cover and maintain ecological balance.
14. **Answer: C) Sandy soil — Holds water strongly**
    Sandy soil does not hold water well; it drains quickly.

15. **Answer: A) Hard rocks**
Both igneous and metamorphic rocks are generally hard and crystalline, unlike sedimentary rocks which may contain fossils.

One or more than one type

1. **Correct Answers:** A) Sandstone, B) Shale, D) Limestone
Sedimentary rocks are formed by the deposition and compaction of sediments over time.
- Sandstone, Shale, and Limestone are common sedimentary rocks.
- Marble is not sedimentary; it is a metamorphic rock formed from limestone under heat and pressure.

2. **Correct Answers: A) Oxygen, B) Silicon**
The Earth's crust mainly consists of oxygen and silicon, which together form most minerals (like silicates).
Oxygen is the most abundant element.
Silicon is the second most abundant.
Iron and carbon are present but in smaller amounts.

3. **Correct Answers: A) Deforestation, B) Overgrazing, C) Urbanization**
Soil erosion happens when the top layer of soil is removed.
Deforestation removes trees that hold soil together.
Overgrazing reduces vegetation cover.
Urbanization disturbs land and exposes soil.
Volcanic eruptions are natural events but not a common major cause of soil erosion.

4. **Correct Answers: A) Air, B) Water, C) Soil, D) Minerals**
All listed options are essential resources:
- Air is needed for breathing.
- Water is vital for all living organisms.
- Soil supports plant growth.
- Minerals are required for body functions and industrial uses.

Assertion and reason type:

1. **Correct Answer: C. Assertion (A) is true and Reason (R) is false.**
Explanation:
- The inner core is solid, even though it has extremely high temperatures.
- This is because of immense pressure at the center of the Earth, which keeps the material in a solid state.
- The reason is false because the temperature of the inner core is actually higher than the outer core, not lower.

2. **Correct Answer: C. Assertion (A) is true and Reason (R) is false.**
Explanation:
- Minerals are indeed naturally occurring inorganic substances found in the Earth’s crust.
- The reason is false because minerals are not formed from plant and animal remains.
- Substances formed from such remains (like coal and petroleum) are fossil fuels, not minerals.

Match the following

1. **Correct Matching**: **(a) → (P);(b) → (Q) ;(c) → (R);(d) → (S)**
   Explanation:

- Igneous rocks form from cooling magma → *Basalt* is a common example.
- Sedimentary rocks form from deposited sediments → *Sandstone* fits this category.
- Metamorphic rocks form due to heat and pressure → *Marble* is formed from limestone.
- Renewable resources can be replenished naturally → *Water* is renewable through the water cycle.

2. **Correct Matching: (a) → (Q) ;(b) → (P) ;(c) → (R) ;(d) → (S)**
   Explanation:

- Water is essential for aquatic organisms → supports aquatic life.
- Loamy soil has a balanced mix of sand, silt, and clay → ideal for crops.
- Tropical rainforests are thick forests found in hot, wet regions.
- Clayey soil retains water well → suitable for rice cultivation and pottery.

Comprehension type:

Passage-1

1. **Correct Answer: B) Heat and pressure**
   Explanation:
   Metamorphic rocks are formed when existing rocks are subjected to high heat and pressure deep inside the Earth. This changes their structure and composition without melting them.
2. **Correct Answer: B) By compression of layers of sand, silt, and clay over millions of years**
   Explanation:
   Sedimentary rocks form when layers of sediments (like sand, silt, and clay) are deposited, compacted, and cemented together over long periods.
3. **Correct Answer: B) Yes, igneous rocks can break down into small pieces that form sedimentary rocks**
   Explanation:
   Igneous rocks can undergo weathering and erosion, breaking into small particles. These particles form sediments, which later become sedimentary rocks. This is part of the rock cycle.

Passage-2

1. **Correct Answer: B) It provides nutrients, water, and support for growth**
   Explanation:
   Soil supplies essential nutrients, holds water, and provides physical support for plant roots, helping plants grow properly.
2. **Correct Answer: B) Increases soil erosion and reduces fertility**
   Explanation:
   Trees hold the soil together with their roots. When forests are cut down, the soil becomes loose and is easily washed away, leading to erosion and loss of nutrients.
3. **Correct Answer: A) Afforestation and planting trees**
   Explanation:
   Planting trees (afforestation) helps bind the soil with roots, reducing erosion and maintaining soil fertility.

3. Food and Health

Worksheet – 1

1. **Correct answer – C**
   Carbohydrates are the body's preferred and most readily available source of fuel for daily activities.
2. **Correct answer – B**
   Chicken is an animal-based product that provides high-quality proteins essential for muscle repair and growth.
3. **Correct answer – B**
   Vitamin A is vital for maintaining the health of the retina (vision) and the epithelial tissues of the skin.
4. **Correct answer – C**
   Unsaturated fats, typically found in plants and fish, help improve blood cholesterol levels and heart health.
5. **Correct answer – B**
   Dairy products like milk and cheese are the primary dietary sources of calcium, which is the main building block of bones.
6. **Correct answer – B**
   Dairy products like milk and cheese are the primary dietary sources of calcium, which is the main building block of bones.
7. **Correct answer – C**
   Whole grains contain the bran and germ layers that provide the indigestible roughage (fiber) needed for healthy digestion.
8. **Correct answer – C**
   Vitamins act as protective nutrients that regulate chemical reactions and help the immune system fight diseases.
9. **Correct answer – C**
   A balanced meal requires fruits and vegetables for protective nutrients and fiber, which are missing from her high-fat, sugary meal.
10. **Correct answer – C**
    Rickets is often placed in overlapping categories because it involves both a vitamin (D) and a mineral (Calcium), while Beriberi is a specific vitamin (B1) deficiency.

Worksheet – 2

1. **Correct answer – C**
   Fermentation is the chemical breakdown of a substance by bacteria, yeasts, or other microorganisms, typically involving effervescence and the giving off of heat.
2. **Correct answer – B**
   Yogurt is produced by the bacterial fermentation of milk, which converts lactose into lactic acid.
3. **Correct answer – B**
   Probiotics are live "good" bacteria and yeasts that are beneficial for the human digestive system.
4. **Correct answer – B**
   Improves digestion The probiotics in fermented foods help maintain a healthy balance of gut bacteria, which aids in breaking down food.

5. Correct answer – B

Spices like turmeric and chili powder are frequently adulterated with harmful synthetic dyes or lead chromate to enhance their color.

6. Correct answer – B

Adding harmful substances to food Adulteration is the legal term for lowering the quality of food by adding inferior or toxic ingredients for profit.

7. Correct answer – B

It can cause health problems Consuming adulterated food can lead to chronic toxicity, digestive disorders, and long-term organ damage.

8. Correct answer – C

Place Cabbage at Step 1 and bacteria and yeast at Step 2. Sauerkraut is made by combining sliced **cabbage** (raw material) with **bacteria and yeast** (fermenting agents) to produce the final product.

9. Correct answer – B

Place "Made from milk" into intersection B and "Contains good bacteria" into intersection Y. Yogurt is specifically milk-based, while the presence of "good bacteria" (probiotics) is a shared characteristic of all fermented foods.

10. Correct answer – C

(C) Path B illustrates the 'Health Risks' and 'Misleading Quality' consequences of adulteration described in the text. The visual pathway showing a person becoming sick highlights how adulterants deceive the consumer and cause immediate or long-term illness.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct answer – B

Proteins are the building blocks of the body, made of amino acids used for growth and tissue repair.

2. Correct answer – B

Iodine is required by the thyroid gland; its deficiency causes swelling in the neck known as goitre.

3. Correct answer – C

Vitamin D helps the body absorb calcium, which is essential for hardening bones and teeth.

4. Correct answer – C

A balanced diet must include carbohydrates, proteins, fats, vitamins, minerals, fiber, and water in the right proportions.

5. Correct answer – B

Iron is a key component of hemoglobin, the protein in red blood cells that carries oxygen.

6. Correct answer – B

Vitamin C deficiency leads to scurvy, characterized by bleeding gums and slow wound healing.

7. Correct answer – A

Bread is made using yeast, which ferments the sugars in flour to produce carbon dioxide, making the dough rise.

8. Correct answer – D

Plastic is a synthetic polymer and not a biological nutrient required for life.

9. Correct answer – C

Starch is often added to milk to increase its viscosity (thickness) and hide the fact that it has been diluted with water.

10. Correct answer – A

Also known as roughage, fiber adds bulk to the diet and helps in the smooth movement of food through the digestive tract.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, B, D

Rice and wheat are grains (complex carbs), and potatoes are starchy tubers. Spinach is primarily a source of vitamins and minerals.

12. Correct answer – A, B, C

A balanced diet needs all these. "Only Fats" is incorrect because fats should be taken in limited quantities alongside other nutrients.

13. Correct answer – A, C

Vitamin D deficiency leads to rickets (weak bones) and can cause muscle pain/weakness. Night blindness is Vitamin A; Scurvy is Vitamin C.

14. Correct answer –A, B, C

Yogurt (milk), Kimchi (vegetables), and Cheese (milk) are all produced via fermentation. Plain rice is usually boiled, though fermented rice products exist (like idli batter).

15. Correct answer – A, B, C

Water, salt, and sand are common physical adulterants used to increase weight/volume. Extra vitamins would be considered "fortification," which is beneficial, not adulteration.

ASSERTION AND REASNING TYPE

16. Correct answer – B

Both are true (Proteins build tissue AND can be used for energy if carbs/fats are low), but the fact that they provide energy is not the *reason* why they are used for growth (that is due to amino acids).

17. Correct answer – B

Both are true (Vit C prevents scurvy AND helps iron absorption), but iron absorption is not the reason it prevents scurvy (it prevents scurvy by helping collagen synthesis).

18. Correct answer – C

Assertion is true, but Reason is false. Fiber cannot be digested by humans and therefore provides almost no energy.

19. Correct answer – A

Both are true. Bones act as a reservoir for calcium; because the body stores it there, calcium is fundamentally necessary for the bone's structure.

MATCH THE FOLLOWING TYPE

20. Correct answer – a-Q, b-S, c-R, d-P

21. Correct answer – a-P, b-S, c-R, d-Q

COMPREHENSHION TYPE

Passage1

22. Correct answer - (B) To help children grow strong and healthy.

23. Correct answer – C A variety of foods.

24. Correct answer – C Carbohydrates.

Passage2

25. Correct answer – B Lack of essential nutrients in the diet.

26. Correct answer – C Vitamin C.

27. Correct answer – B Bleeding gums.

Previous year questions

1. Correct answer – C

Gives energy instantly – Glucose is a simple sugar that does not need complex digestion; it enters the bloodstream quickly to provide immediate fuel.

2. Correct answer – B

Potassium must be changed to phosphorus – Bones are primarily composed of Calcium and Phosphorus (as hydroxyapatite). Potassium is an intracellular fluid mineral.

3. Correct answer – C

a-(iii), b-(v), c-(ii), d-(iv), e-(i)

Vitamins (Fight disease), Roughage (Faeces removal), Carbs (Energy), Proteins (Muscles), Fats (Warmth).

4. Correct answer – B

In option B, if "Vegetable oil" was the odd one out among Cheese and Milk (which both have protein/calcium), it is correctly identified as a pure fat.

5. Correct answer – C

(a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

Goitre (Iodized salt), Rickets (Fish/Egg/Mushroom for Vit D), Beriberi (Meat/Green veg for Vit B1), Anaemia (Apples/Leafy veg for Iron)

4. Diseases

Worksheet -1

1. **Correct answer – A**
   Tuberculosis (TB) is an infectious disease caused specifically by the bacterium *Mycobacterium tuberculosis*.
2. **Correct answer – A**
   Bacteria are prokaryotic because they lack a defined nucleus and membrane-bound organelles, unlike fungi or parasites.
3. **Correct answer – A**
   Bacteria are ubiquitous (found everywhere) and can survive in extreme environments where other organisms cannot.
4. **Correct answer – D**
   By definition, a parasite lives in or on a host organism and gets its food at the expense of its host.
5. **Correct answer – B**
   Chickenpox is a highly contagious disease caused by the varicella-zoster virus
6. **Correct answer – A**
   Asthma is a non-communicable (lifestyle or genetic) disease, meaning it cannot be spread from one person to another like the flu or ringworm.
7. **Correct answer – A**
   Eating contaminated food or water often leads to digestive distress, such as diarrhoea, as the body tries to flush out toxins or germs.
8. **Correct answer – D**
   Seeking medical advice early, when symptoms persist, or when you are uncertain helps prevent minor issues from becoming serious.
9. **Correct answer – D**
   Heart disease is a "lifestyle disease" typically caused by long-term unhealthy habits, whereas the other options are caused by germs.
10. **Correct answer – B**
    A fever is a protective biological response where the body raises its temperature to make the environment less "friendly" for invading germs.

Worksheet – 2

1. **Correct answer – B**
   Pneumonia is an infection that inflames the air sacs in one or both lungs, directly affecting the respiratory system.
2. **Correct answer – C**
   Malaria is caused by the *Plasmodium* parasite, which is transmitted to humans through the bite of an infected female Anopheles mosquito.
3. **Correct answer – B**
   Bacteria in the mouth turn sugar from food into acid, which eats away at tooth enamel and causes decay.
4. **Correct answer – A**
   A fever is a general sign that the body is fighting off an invading germ or infection.
5. **Correct answer – B**
   Typhoid fever is a bacterial infection (Salmonella Typhi) spread specifically through the "fecal-oral" route of contaminated food and water.

6. Correct answer – C
Place Bandages with "Cover small cuts and scrapes" and Tweezers with "Remove splinters". This correctly identifies the basic first-aid function of these tools.

7. Correct answer – C
Tell an adult immediately. In any emergency sequence involving a child, the most critical step after identifying an injury is alerting a responsible adult.

8. Correct answer – B
Always tell an adult immediately. Alerting an adult is a core safety rule that applies to both medical interventions (vaccinations) and immediate first aid.

9. Correct answer – D
Cover your mouth and nose with your elbow when you cough or sneeze. This "elbow cover" prevents germs from traveling through the air to other people.

10. Correct answer – D
In still water Mosquitoes require stagnant or "still" water (like in puddles or buckets) to lay their eggs and grow larvae.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct answer – B
The common cold is highly contagious and spreads through air droplets.

2. Correct answer – A
Because it is passed from one person to another via a mosquito, it is considered communicable.

3. Correct answer – C
Hypertension (high blood pressure) is a lifestyle disease and cannot be "caught" from someone else.

4. Correct answer – B
Typhoid is caused by *Salmonella typhi* bacteria; Malaria is protozoan and Flu is viral.

5. Correct answer – C
Vaccinations prepare the immune system to fight specific germs before they can cause illness.

6. Correct answer – C
Mosquitoes act as "vectors" that carry the malaria parasite from person to person.

7. Correct answer – B
Cancer is caused by uncontrolled cell growth, not by a transmissible germ.

8. Correct answer – B
Washing removes dirt and germs to prevent infection before applying a bandage.

9. Correct answer – C
Most colds are caused by "rhinoviruses," which are a type of virus.

10. Correct answer – B
Washing hands disrupts the transmission of viral particles like the flu from surfaces to your face.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – (A, C) Influenza and TB are communicable; Asthma and Heart disease are non-communicable.

12. Correct answer – (A, B, D) These three directly stop the transfer of germs; exercise improves health but doesn't stop germ spread.

13. **Correct answer – (A, C)** Malaria and Dengue are both spread by mosquitoes (insects).
14. **Correct answer – (A, B, D)** Salmonella (bacteria), contaminated water, and viruses (like Norovirus) all cause foodborne illness.
15. **Correct answer – (A, B, D)** High blood sugar is a symptom of diabetes, not a respiratory cold.

ASSERTION AND REASNING TYPE

16. **Correct answer – C**
A is true, but R is false (Malaria is spread by mosquitoes, not food/water).
17. **Correct answer – A**
Both are true, and R is the direct reason why A works.
18. **Correct answer – D**
A is false (Cancer is generally not viral), and R is true (it is non-communicable).
19. **Correct answer – A**
Both are true; antibiotics are biologically designed to kill bacteria, not viruses.

MATCH THE FOLLOWING TYPE

20. **Correct answer –** (a)-Q, (b)-R, (c)-S, (d)-P
21. **Correct answer –** (a)-Q, (b)-P, (c)-S, (d)-R

COMPREHENSHION TYPE

Passage1

22. **Correct answer - (B)** Flu
23. **Correct answer - (B) Sneezing**
24. **Correct answer - (B) Sore throat**

Passage2

25. **Correct answer - (B) John**
26. **Correct answer - (B) Mosquito**
27. **Correct answer - (C) One week**

Previous Year Questions

1. **Correct answer – B**
Digestion begins in the mouth, continues in the stomach, and is **completed in the small intestine** where nutrients are absorbed.
2. **Correct answer – B**
Direct contact – Diabetes is incorrect because diabetes is a non-communicable metabolic disorder.
3. **Correct answer – C**
R can be plague is correct; Plague is a bacterial disease spread to humans by the bite of infected rat fleas.
4. **Correct answer – A**
Malaria of group I should be interchanged with typhoid of group III. *Group I (Bacteria):* TB and Tetanus are bacterial. Malaria (Protozoa) is the odd one. *Group III (Protozoa):* Kala azar and Sleeping sickness are protozoan. Typhoid (Bacteria) is the odd one.
5. **Correct answer – C**
3 The diseases from the list spread by contaminated food/water are **Diarrhoea, Cholera, and Typhoid**. (Flu/Measles are air; Dengue/Plague are insects; Ringworm is contact).

5. Plants Around Us

Worksheet – 1

1. **Correct answer – C**
   Roots grow underground to anchor the plant and act like "straws" to suck up water and minerals from the soil.
2. **Correct answer – C**
   The stem provides the structural "backbone" for the plant and contains tubes (xylem and phloem) to transport water and food.
3. **Correct answer – B**
   This is the magical "cooking" process where plants use chlorophyll to turn sunlight, water, and carbon dioxide into energy.
4. **Correct answer – B**
   Petals are the "billboards" of the flower, using bright colors and sweet smells to tell bees and butterflies, "Come visit me!"
5. **Correct answer – C**
   The ovary is the protective chamber at the base of the flower where seeds grow; it often turns into a fruit later!
6. **Correct answer – C**
   The style is the long, slender stalk that acts as a "bridge" or tube for pollen to travel down to reach the ovary.
7. **Correct answer – B**
   The receptacle is the thickened base of the flower that acts like a stage, holding all the other floral parts in place.
8. **Correct answer – B**
   The stamen is the male part, made of the anther (which makes the yellow pollen "dust") and the filament.
9. **Correct answer – C**
   Sepals are the tough, leaf-like structures that wrap around the flower bud to keep it safe from bugs and weather before it opens.
10. **Correct answer – C**
    As the "cozy blanket," the green sepal is the first line of defense for a developing flower bud.

Worksheet – 2

1. **Correct answer – B**
   Sexual reproduction Flowering plants primarily use seeds, which are formed through the fusion of male and female gametes, to reproduce.
2. **Correct answer – C**
   Stamen The stamen is the male part of the flower, and its top part (the anther) is specifically responsible for making pollen.
3. **Correct answer – C**
   Rose Roses are classic examples of bisexual flowers because a single bloom contains both the stamen (male) and the pistil (female).
4. **Correct answer – B**
   By transferring pollen from one flower to another These insects act as "delivery drivers," carrying pollen on their bodies so that fertilization can occur.

5. **Correct answer – B**
A seed is formed in the ovary Once the pollen reaches the ovule, fertilization occurs, turning the ovule into a seed and the ovary into a fruit.
6. **Correct answer – B**
Take stem cuttings and plant them in moist soil This is a form of vegetative propagation that creates identical "clones" of the parent plant much faster than growing from seeds.
7. **Correct answer – B**
Contains both male and female parts The term "bisexual" in botany means the flower possesses both a stamen and a pistil in the same structure.
8. **Correct answer – C**
To produce pollen the anther is the specialized "factory" at the tip of the stamen where pollen grains are developed and released.
9. **Correct answer – C**
Tuber The "eyes" on a potato tuber are actually buds that can sprout and grow into entirely new potato plants.
10. **Correct answer – B**
To make new plants quickly Vegetative propagation allows plants to bypass the slow seed-growing process and produce offspring that are exactly like the parent.

Worksheet – 3

1. **Correct answer – B**
Pollination is the physical transfer of pollen from the anther to the stigma.
2. **Correct answer – C**
The **Stamen** (specifically the anther) produces pollen.
3. **Correct answer – B**
Self-pollination occurs when pollen from the same flower or plant fertilizes the stigma.
4. **Correct answer – B**
Dispersal prevents overcrowding (competition) around the parent plant.
5. **Correct answer – B**
Corn is a **monocot** (1 cotyledon) and Gram is a **dicot** (2 cotyledons).
6. **Correct answer – B**
By moving pollen as they feed on nectar.
7. **Correct answer – B**
Germination is the "waking up" of the seed using water and warmth.
8. **Correct answer – B**
Pea plants use pods that "pop" or explode to scatter seeds.
9. **Correct answer – C**
Female flowers only have a pistil, so they **must** receive pollen from outside.
10. **Correct answer – B**
They cannot self-pollinate easily, forcing them to mix genes with other plants.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C**
Leaves contain chlorophyll, which captures sunlight to produce food during photosynthesis.

2. Correct answer – B
Roots serve as the primary organ for absorbing water and essential minerals from the soil.

3. Correct answer – B
Xylem is the specialized vascular tissue that transports water and minerals from the roots upwards.

4. Correct answer – C
Flowers contain the reproductive organs necessary for the production of seeds and fruits.

5. Correct answer – B
Many plants use modified roots (like carrots or sweet potatoes) to store energy in the form of starch.

6. Correct answer – B
Transpiration, the evaporation of water from plants, occurs mainly through tiny pores in the leaves called stomata.

7. Correct answer – B
The stem acts as a structural pillar and a transport highway for moving water and nutrients.

8. Correct answer – C
Phloem is the vascular tissue responsible for the translocation of sugars from the leaves to the rest of the plant.

9. Correct answer – C
Bud scales or sepals provide a protective outer layer for the delicate developing buds.

10. Correct answer – D
Flowers are the site of sexual reproduction, housing the male stamens and female pistils.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – (A)
Only roots are primarily responsible for the uptake of water and nutrients from the earth.

12. Correct answer – (A, B)
Leaves are the sites for both food production (photosynthesis) and breathing (gas exchange).

13. Correct answer – (A, B)
Xylem and Phloem together form the vascular system that moves substances throughout the plant.

14. Correct answer – (A, D)
The stamen (male) and pistil (female) are the two primary reproductive organs of a flower.

15. Correct answer – (A, C, D)
Stems provide support, transport materials, and can store food (like in potato tubers or ginger).

ASSERTION AND REASNING TYPE

16. Correct answer – B
Both statements are facts, but root absorption is not the reason *why* leaves perform photosynthesis.

17. Correct answer – B
Both are true; while transport is the main job, some green stems (like cacti) also produce food.

18. Correct answer – B
Both are true statements about different organs, but leaf storage does not explain floral reproduction.

19. Correct answer – D
Both are false because Xylem transports water (not sugar) and Phloem transports sugar (not water).

MATCH THE FOLLOWING TYPE

20. Correct answer – (a)-Q, (b)-P, (c)-R, (d)-S

21. Correct answer – (a)-P, (b)-Q, (c)-R, (d)-S

COMPREHENSHION TYPE

Passage1

22. Correct answer – B Plants are primary producers that convert sunlight into chemical energy.

23. Correct answer – B Plants improve city life by filtering out harmful air pollutants.

24. Correct answer – C The loss of plants would collapse the food chain and destroy ecological stability.

Passage2

25. Correct answer – B Trees provide natural cooling (shade), which significantly lowers the need for electricity.

26. Correct answer – B Green environments created by trees help lower stress and improve our overall quality of life.

27. Correct answer – B Cities use tree-planting initiatives to create more green spaces and engage the community.

Previous Year Questions

1. Correct answer – B
Tigernail (or Martynia) has hooked fruits designed to latch onto animals for dispersal.

2. Correct answer – A
Rice (Monocot) – Pea (Dicot) Cereal grains like rice/wheat have one cotyledon; legumes like peas/beans have two.

3. Correct answer – D
S In a seed diagram, **S** usually points to the **Cotyledon**, which stores food for the embryo.

4. Correct answer – A
ii) and (iii) only Sweet potatoes are "modified roots" that store food and absorb water; they do not give color or transport specifically to leaves only.

5. Correct answer – B
(STALK) Unscrambling the letters gives **STALK** (also known as the petiole), which attaches the leaf blade to the stem.

6. Animals Around us

Worksheet – 1

1. **Correct answer – B**
   A habitat is the specific environment where a plant or animal naturally lives, eats, and grows.
2. **Correct answer – B**
   "Terrestrial" means land-based, and forests are one of the most common land habitats on Earth.
3. **Correct answer – B**
   These are the "Big Three" survival essentials that every living thing must get from its surroundings.
4. **Correct answer – C**
   Rivers and oceans Aquatic habitats are water-based environments, including both freshwater (rivers) and saltwater (oceans).
5. **Correct answer – C**
   Dolphins are marine mammals perfectly adapted for life in the saltwater of the world's oceans.
6. **Correct answer – B**
   Deserts are defined by their dry conditions, which limit the types of plants that can grow there.
7. **Correct answer – C**
   An animal's habitat is like a customized survival kit that matches its specific physical needs.
8. **Correct answer – C**
   The camel's ability to store energy and conserve water is a specialized "evolutionary upgrade" for desert survival.
9. **Correct answer – C**
   Polar regions, like the Arctic and Antarctic, are the coldest places on Earth where only specialized life can survive.
10. **Correct answer – B**
    Grasslands are vast, open areas where the rainfall is enough for grass but not quite enough for thick forests.

Worksheet – 2

1. **Correct answer – B**
   Adaptation is all about **survival**- it's the toolkit organisms use to stay alive in tricky environments.
2. **Correct answer – D**
   Structural refers to a physical body part; a giraffe's long neck is a built-in "ladder" for reaching food.
3. **Correct answer – C**
   Behavioural adaptations are actions, like birds flying south or bears sleeping through winter.
4. **Correct answer – B**
   Camels are the ultimate desert pros because they can store fat/water to last through long, dry journeys.

5. Correct answer – B
Whether it's finding food or avoiding becoming food, adaptation is the key to staying in the game of life.

6. Correct answer – A
Energy flows from the **Producer** (Algae) up to the top predator (Kingfisher) in a logical "who eats whom" order.

7. Correct answer – B
Gills are a fish's most important structural adaptation for "breathing" oxygen dissolved in water.

8. Correct answer – A
In a dense forest, having **strong limbs** to swing and climb is much better than having a long neck!

9. Correct answer – C
Both are **Structural**; they are physical body parts (fur and fat) that the bear is born with to handle the cold.

10. Correct answer – B
Blubber acts like a heavy-duty winter jacket, trapping body heat inside where it belongs.

Worksheet – 3

1. Correct answer – B
Lions are carnivores (**Sharp teeth**) and mammals (**Lungs**).

2. Correct answer – C
Sharks use **fins** to balance and steer through the water.

3. Correct answer – B
Herbivores (like cows) have flat teeth for grinding grass.

4. Correct answer – C
Bears are classic omnivores—they'll eat berries and fish!

5. Correct answer – B
Gills pull oxygen directly out of the water as it passes through.

6. Correct answer – C
Frogs are dual-threats: they **jump** on land and **swim** in water.

7. Correct answer – B
Jellyfish (and octopuses!) use pulses of water to push forward.

8. Correct answer – B
A cheetah's speed comes from its **specialized skeleton** and powerful legs.

9. Correct answer – B
Moving fast burns a lot of "fuel" (food and oxygen).

10. Correct answer – C
Herbivores have digestive systems built specifically for **plant matter**.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer – C
While **Hibernation** is a winter nap to avoid cold, **Aestivation** is a summer nap to stay moist and cool during extreme heat.

2. Correct answer – A
Goats have flat teeth for grinding grass. Sharks, tigers, and eagles are built for hunting and eating meat (Carnivores).

3. Correct answer – C
Thinking and sleeping are mental or physical states, but **Flying** is a specific way animals get from point A to point B.

4. Correct answer – B
Carnivores eat other animals. Fish are a great source of protein for predators like bears or larger fish.

5. Correct answer – C
This is a defining trait of mammals. Most don't lay eggs (except the Platypus!), they use lungs, and they are warm-blooded.

6. Correct answer – B
Movement is a survival tool. If you can't move to your dinner—or move away from someone who wants *you* for dinner—life gets very difficult!

7. Correct answer – B
Feathers provide the lift and streamlined shape needed for flight.

8. Correct answer – B
Most land animals need to extract oxygen from the air. Gills are for breathing in water, and fins are for steering.

9. Correct answer – C
Lions are "Apex Predators," meaning they are at the top of the food chain and eat only meat.

10. Correct answer – B
Since it eats only leaves (Herbivore) and flits from plant to plant (using wings), this is the most accurate classification.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – (A, B, C, D) All these traits—fat storage, protective eyelashes, long legs, and thick fur—are specialized "desert upgrades" for the camel.

12. Correct answer – (B, C, D) Snakes, birds, and adult frogs use lungs to breathe air; most fish use gills to extract oxygen from water.

13. Correct answer – (B, C, D) Herbivores are strictly plant-based eaters, consuming everything from fruits and leaves to the entire plant.

14. Correct answer – (B, C, D) Frogs, kangaroos, and rabbits have powerful hind legs designed for jumping, while elephants are too heavy to lift all four feet off the ground at once.

15. Correct answer – (B, C, D) Lightweight bodies, wings for lift, and hollow bones are the three essential engineering feats that allow birds to fly.

ASSERTION AND REASNING TYPE

16. Correct answer – C
The Assertion is true, but the Reason is false because movement is absolutely vital for an animal to find resources and stay alive.

17. Correct answer – C
The Assertion is true, but the Reason is false because gills are designed to extract oxygen from **water**, not air.

18. Correct answer – C
The Assertion is true, but the Reason is false because herbivores have flat, grinding teeth for plants, not sharp teeth for meat.

19. Correct answer – C
The Assertion is true, but the Reason is false because different habitats (like the deep sea vs. a desert) require very different types of adaptations.

MATCH THE FOLLOWING TYPE

20. Correct answer – (a-R, b-P, c-Q, d-S)

21. Correct answer – (a-P, b-Q, c-R, d-S)

COMPREHENSHION TYPE

Passage1

22. Correct answer – B Blubber and fur act as a thermal blanket for polar bears.

23. Correct answer – A Water and fat storage allow camels to survive long periods without fresh resources.

24. Correct answer – C Gills are the specialized "underwater lungs" for fish.

Passage2

25. Correct answer – B Movement is a survival requirement for finding food, mates, and safety.

26. Correct answer – C Wings provide the surface area and power needed for avian flight.

27. Correct answer – B Fins provide the stability and propulsion needed for swimming.

Previous Year Questions

1. Correct answer – B
X and V: Based on standard NSO classifications, **X** represents animals living in both water/land (Newt), and **V** represents egg-laying mammals (Spiny Anteater).

2. Correct answer – B
Eel: Group B consists of terrestrial (land) animals, but the **Eel** is an aquatic animal and belongs in Group A.

3. Correct answer – D
Both B and C: The animal described (Locust) is famous for traveling in massive, destructive swarms across continents.

4. Correct answer – D
Prawn & Housefly: P represents an aquatic animal breathing through gills (Prawn), and **Q** represents a land-based insect with wings (Housefly).

5. Correct answer – C
1 and 3: All food chains must begin with a producer (plant), and those who eat the plants (herbivores) eventually become energy for predators.

7. Human Body

Worksheet – 1

1. **Correct answer – B**
   Muscles are designed to contract and relax, which pulls on our bones to create movement.
2. **Correct answer – C**
   Skeletal muscles are "voluntary," meaning you consciously decide to move them, like when you wave hello.
3. **Correct answer – C**
   These muscles are attached to the skeleton by tendons and work like pulleys to move our limbs.
4. **Correct answer – C**
   This is a unique, involuntary muscle that never gets tired and keeps your heart pumping 24/7.
5. **Correct answer – B**
   Muscle fatigue is the body's way of saying it needs a break to clear out waste products and recover energy.
6. **Correct answer – C**
   Think of **L**igaments as the "Links" that hold **B**one to **B**one at your joints.
7. **Correct answer – C**
   The skeleton provides the sturdy framework that holds us up and protects our soft internal organs.
8. **Correct answer – B**
   The human skull is a puzzle made of 22 bones: 8 that protect the brain and 14 that form your face.
9. **Correct answer – C**
   This joint works like a joystick, allowing the most freedom of movement in almost every direction.
10. **Correct answer – B**
    The spine is made of 33 small, ring-like bones called vertebrae that protect your spinal cord.

Worksheet – 2

1. **Correct answer – B**
   To transport oxygen, nutrients, and waste throughout the body Think of it as a massive delivery service that visits every single cell in your body.
2. **Correct answer – D**
   The heart is a powerful muscle that never stops "pushing" blood through your system.
3. **Correct answer – C**
   A simple way to remember: **A**rteries carry blood **A**way from the heart.
4. **Correct answer – A**
   To carry oxygen-poor blood back to the heart Once the cells take the oxygen, veins act as the "return lane" to send blood back for a refill.
5. **Correct answer – B**
   Tiny blood vessels where the exchange of oxygen and nutrients occurs These are so thin that nutrients can actually "leak" through the walls to feed your cells.

6. **Correct answer – C**
   The brain is the "Boss"—it processes every thought, memory, and movement you have.
7. **Correct answer – C**
   The spinal cord is the main "data cable" protected inside your backbone.
8. **Correct answer – D**
   Your nerves feel the heat, your spinal cord manages the reflex, and your brain realizes, "Ouch!"
9. **Correct answer – B**
   Nerves are like electrical cables that send signals at lightning speed across your body.
10. **Correct answer – B**
    It acts as the bridge that connects your "Main Computer" (brain) to the rest of your "Hardware" (body).

Worksheet – 3

1. **Correct answer – B**
   5 – Humans have five primary senses: sight, hearing, smell, taste, and touch.
2. **Correct answer – C**
   Eyes – The eyes act as cameras that capture light and send visual data to the brain.
3. **Correct answer – C**
   Lens – The lens is a transparent structure that changes shape to focus light perfectly onto the retina.
4. **Correct answer – A**
   Stirrup bone – Located in the middle ear, the stapes (stirrup) is roughly the size of a grain of rice.
5. **Correct answer – C**
   They help us taste different flavours – Taste buds detect chemical signals for sweet, sour, salty, and bitter.
6. **Correct answer – C**
   Nose – The nose contains specialized receptors that detect chemical molecules in the air.
7. **Correct answer – B**
   To turn sound vibrations into signals – The cochlea is a fluid-filled spiral that converts mechanical waves into electrical nerve impulses.
8. **Correct answer – C**
   Skin – The skin covers the entire exterior of the body, making it the largest organ by surface area and weight.
9. **Correct answer – C**
   Different sensations – Nerve endings act as sensors for external stimuli like pressure, temperature, and pain.
10. **Correct answer – B**
    Nose – Olfactory cells are located in the upper part of the nasal cavity to detect scents.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C**
   Nose – Olfactory cells are located in the upper part of the nasal cavity to detect scents.
2. **Correct answer – C**
   Breathe oxygen – Lungs facilitate the exchange of oxygen and carbon dioxide.
3. **Correct answer – A**
   Nervous system – It acts as the body's command center, managing all voluntary and involuntary actions.
4. **Correct answer – C**
   Mouth – Digestion begins mechanically (teeth) and chemically (saliva) in the mouth.
5. **Correct answer – A**
   Protect organs – The skeleton provides a rigid cage to shield delicate organs like the brain and heart.
6. **Correct answer – C**
   Alveoli – These tiny sacs are where oxygen actually enters the bloodstream.
7. **Correct answer – B**
   Four – The human heart has two atria and two ventricles to keep oxygen-rich and oxygen-poor blood separate.
8. **Correct answer – C**
   Stomach – The stomach uses churning and acid to turn food into a semi-liquid state.
9. **Correct answer – B**
   Brain – The brain is the primary organ of the central nervous system.
10. **Correct answer –C**
    Tendons – **T**endons **T**ie **M**uscle to **B**one.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – (A, B, D)** The skeleton protects organs, stores calcium/phosphorus, and provides levers for movement.
12. **Correct answer – (A, C)** The liver produces bile and the stomach uses acid; both are vital for digestion.
13. **Correct answer – (A, B, C, D)** The nervous system coordinates everything from survival (breathing) to complex logic (thinking).
14. **Correct answer – (A, B, D)** These three types of vessels form the "piping" for blood flow.
15. **Correct answer – (A, D)** The heart pumps blood, which is the vehicle that supplies oxygen to all cells.

ASSERTION AND REASNING TYPE

16. **Correct answer – D**
    Assertion is false (Brain is nervous system), but Reason is true (it does control responses).
17. **Correct answer – A**
    Both are true, and the need for oxygen at a cellular level is exactly why we have lungs.

18. Correct answer – D
Assertion is false (Heart pumps blood, not stomach), but Reason is true.

19. Correct answer – C
Assertion is true (Bones store calcium), but Reason is false (Bones are skeletal, not muscular).

MATCH THE FOLLOWING TYPE

20. Correct answer – (a-R, b-P, c-Q, d-S)

21. Correct answer – (a-R, b-S, c-Q, d-P)

COMPREHENSHION TYPE

Passage1

22. Correct answer – C
Beats faster to meet the body's increased demand for resources.

23. Correct answer – A
To deliver oxygen and nutrients to working muscles more quickly.

24. Correct answer – B
Oxygen is the "fuel" needed for the body to produce energy.

Passage2

25. Correct answer – B
Its primary job is converting food into usable energy (nutrients).

26. Correct answer – C
Gastric acid in the stomach breaks down complex food particles.

27. Correct answer – B
The small intestine has a massive surface area specifically for nutrient absorption.

Previous Year Questions

1. Correct answer – A
Incorrect; the **Retina** is the light-sensitive layer, whereas the Cornea is the clear outer protective layer.

2. Correct answer – C
Z, Y – The pelvic girdle is part of the appendicular skeleton (Y), and the spinal cord is part of the central nervous system (Z).

3. Correct answer – A
Shoulder – A joystick move is a 360° rotation, which is only possible in a **Ball-and-Socket** joint like the shoulder.

4. Correct answer – B
(i) Ball-and-Socket (Circle), (ii) Hinge (Kneel), (iii) Gliding (Wrist), (iv) Pivot (Neck).

5. Correct answer – C
Cardiac, Skeletal, Smooth – X is involuntary/striated (Heart), Y is voluntary/striated (Arm/Leg), Z is involuntary/non-striated (Stomach).

8. Work and Energy

Worksheet – 1

1. **Answer: B) Energy**
   Energy is scientifically defined as the ability or capacity to do work.
2. **Answer: C) Kicking a football**
   Work only happens when a force causes an object to move. Only the football moves in these examples.
3. **Answer: C) Electrical Energy**
   These devices require electric currents to function.
4. **Answer: C) A ringing bell**
   A ringing bell generates sound energy through vibrations, not heat.
5. **Answer: C) Sun**
   The Sun's energy drives the climate and allows plants to make food via photosynthesis.
6. **Answer: C) Light Energy**
   Our eyes detect light reflecting off objects, enabling vision.
7. **Answer: B) A burning candle**
   The chemical reaction of burning produces both heat and light.
8. **Answer: C) Electrical Energy**
   Refrigerators plug into wall outlets to use electricity for their cooling compressors.
9. **Answer: A) Sound Energy**
   Clapping creates vibrations in the air that travel to our ears as sound.
10. **Answer: C) Heat Energy**
    Heat must be added to water to raise its temperature until it boils.

Worksheet – 2

1. **Answer:** C) Electrical energy into sound energy:
   A radio uses electricity to power speakers that produce sound waves.
2. **Answer:** D) Coal:
   Coal is a finite fossil fuel that takes millions of years to form.
3. **Answer:** C) Petroleum
   Like coal, it is a fossil fuel that cannot be easily replenished once extracted.
4. **Answer:** C) They cause pollution
   Burning non-renewable fossil fuels releases greenhouse gases and harmful emissions.
5. **Answer:** B) Wind Energy:
   Wind turbines capture the kinetic energy of moving air to generate electricity.
6. **Answer:** A) Biomass Energy:
   This energy is derived from organic materials like plant and animal waste.
7. **Answer:** B) Coal:
   It is a fossil fuel mined from the earth and is not naturally replenished on a human timescale.
8. **Answer:** B) Water:
   Hydroelectric plants rely on the flow of water to turn turbines.
9. **Answer:** C) Solar Energy:
   It is a clean, renewable resource that does not emit pollutants.

10. **Answer:** B) Nuclear Energy:
It relies on uranium, a finite element mined from the Earth, making it non-renewable.

Competitive Corner

Single correct option Type

1. **Answer:** B) Bent wooden bow:
The bending of the bow stores elastic potential energy, which transfers to the arrow as kinetic energy.

2. **Answer:** D) Weight energy:
Weight is a measurement of gravitational force, not a distinct form of energy.

3. **Answer:** C) Heat energy:
The friction from rubbing hands together converts mechanical movement into heat.

4. **Answer:** A) A rolling ball:
Mechanical energy encompasses the energy of moving objects (kinetic energy).

5. **Answer:** B) Sound energy:
Vibrating objects create sound waves that travel through the air.

6. **Answer:** C) Wind:
Wind is constantly generated by the sun's uneven heating of the Earth, making it infinitely renewable.

7. **Answer:** B) Natural gas:
This fossil fuel is commonly piped into homes for cooking and heating.

8. **Answer:** B) They cause pollution:
Combusting fossil fuels releases harmful pollutants into the atmosphere.

9. **Answer:** D) Mechanical energy into sound energy:
Physically striking or shaking the bell causes it to vibrate and ring.

10. **Answer:** B) Potential energy:
Stretching a rubber band stores elastic potential energy based on its changed shape.

11. **Answer:** C) Water:
Hydroelectric dams use the kinetic energy of flowing water to generate power.

12. **Answer:** B) Electrical energy changes to light energy:
A battery supplies electrical current to illuminate the torch's bulb.

13. **Answer:** C) Natural gas:
It is a fossil fuel extracted from underground and cannot be replaced once used.

14. **Answer:** C) Petroleum:
Most vehicles use fuels like gasoline and diesel, which are refined from crude oil (petroleum).

15. **Answer:** A) Mechanical energy:
Pedaling a bicycle is a physical, mechanical action that moves the wheels.

ONE OR MORE THAN CORRECT OPTION TYPE

16. **Correct Options: A, B, C, and D**
Sound energy is the energy produced by vibrating objects and carried through waves.
Heat energy (thermal energy) is the energy arising from the internal motion of atoms and molecules.
Electrical energy is the energy generated by the movement of electric charges.

Potential energy is the stored energy an object possesses due to its position or state.

17. Correct Options: A and B

Wind energy and Solar energy are renewable because they are naturally replenished on a human timescale and will never run out.

Petroleum and Coal are fossil fuels. They are non-renewable because they take millions of years to form and are consumed much faster than they can be naturally replaced.

18. Correct Options: A, B, and D

They cause pollution (A): Burning fossil fuels releases harmful greenhouse gases and pollutants into the atmosphere.

They can run out (B): They are exhaustible, finite resources.

They take millions of years to form (D): Because their formation rate is incredibly slow, they cannot be replenished once exhausted, making them unsustainable for long-term use.

19. Correct Options: A, B, and C

A Guitar (vibrating strings) and Drums (vibrating membrane) are musical instruments specifically designed to produce sound energy.

Running water creates vibrations through turbulence and splashing, which generates noticeable sound energy (like the sound of a river or a running tap).

A standard Light bulb is designed to produce light and heat energy, not sound.

20. Correct Options: A, B, and D

A Television, a Washing machine, and a Mobile phone (which runs on a rechargeable battery filled with electrical energy) all require electricity to function.

A Candle relies entirely on chemical energy stored in the wax to produce light and heat; it does not use electricity.

Assertion and Reason Type

21. Answer: C

The assertion is correct because windmills (turbines) harness wind to generate electricity. The reason is false because wind is a naturally replenished, renewable resource.

22. Answer: B

Both statements are factually true. However, the reason does not scientifically explain the assertion; the sun providing clean energy does not explain the mechanical/chemical process (photovoltaic effect) of how a solar panel actually converts that sunlight into electricity.

Matrix Matching Type

23. Answers: A - S, B - Q, C - R, D - P

* Solar energy (A) is harnessed by solar panels to produce electricity (S).

Wind energy (B) is harnessed by windmills to produce electricity (Q).

Biomass energy (C) includes organic matter used to create biogas for cooking (R).

Petroleum (D) is a fossil fuel refined into gasoline/diesel for running vehicles (P).

24. Answers: A - R, B - S, C - P, D - Q

Mechanical Energy (A) is associated with motion and position, like a moving ball or dog (R).

Heat Energy (B) is thermal energy that makes things hot (S).

Light Energy (C) is electromagnetic radiation that helps us see (P).

Energy (D) is scientifically defined as the ability to do work (Q).

Comprehension Type

Passage-I

25. Answer: A

The first passage explicitly states, "Electricity is produced from various sources such as coal, water, wind, and solar energy."

26. Answer: B

The first passage directly supports this, stating electricity is important because "It powers lights, fans, refrigerators, and other appliances."

27. Answer: B

The passage contrasts polluting sources with clean ones. Wind and solar energy are naturally replenished (renewable) and do not emit pollutants (clean).

Passage-II

28. Answer: B

The second passage directly defines why they are non-renewable, stating "these fuels take millions of years to form and are getting used up quickly."

29. Answer: C

According to the second passage, scientists help conserve energy "by looking for alternative energy sources such as solar, wind, and hydro energy."

30. Answer: B

The second passage explicitly lists "solar, wind, and hydro energy" as the alternative energy sources being explored to save fossil fuels.

Previous year questions

1. Answer: B) Force is applied and the object moves

In physics, work is done only when a force acts on an object and the object undergoes displacement in the direction of the force. If there is no movement, no work is done.

2. Answer: B) Joule

The SI unit of work is the **Joule (J)**. One joule of work is done when a force of one newton moves an object through a distance of one metre in the direction of the force.

3. Answer: B) Chemical energy → Kinetic energy

The battery stores **chemical energy**, which is converted into **electrical energy** and then into **kinetic energy** as the car moves.

4. Answer: B, C, D and E

- **A.** Gravitational potential energy is maximum at point **P**. → **Correct**
- **B.** Kinetic energy is maximum at point **P**. → **Incorrect** (it is minimum at the highest point)
- **C.** The toy car uses up all its energy at point **U**. → **Incorrect** (energy is conserved)
- **D.** Kinetic energy is minimum at point **R**. → **Incorrect** (it is maximum at the lowest point)
- **E.** The car has the most kinetic energy at **U**. → **Incorrect** (The car would reach its maximum speed (and most kinetic energy) at the first lowest point (Q). By the time it reaches U, friction would have caused it to lose some mechanical energy)

Therefore, the incorrect statements are **B, C, D and E**.

5. Answer: C) Lifting a heavy suitcase to a higher shelf

Work is done when a force causes displacement. In options **A, B, and D**, there is either no movement or no displacement in the direction of the applied force, so the

work done is zero. In **option C**, the suitcase is moved upward through a distance against gravity, so work is done.

Formula:

$W = F \times d$

where:

W = Work done

F = Force applied

d = Displacement in the direction of the force

Since both force and displacement are involved in option **C**, it represents the maximum work done.

---

9. Force and Simple Machines

Worksheet – 1

1. A) Move toward

Push means applying force to move an object away from you, while pull means applying force to bring an object closer (toward you). These are opposite effects of force.

2. B) Changing the shape of objects

Force can change the **shape, size, speed, or direction** of an object. For example, pressing clay or stretching a rubber band changes its shape. Colour, smell, and temperature are not direct effects of force.

3. D) X-Pulling force, Y- Gravitational force, Z- Frictional force

Arrow X represents Amit's **pulling force**, Y is the downward **gravitational force**, and Z is the opposing **frictional force**,

4. B) Frictional force

When brakes are applied, friction is created between brake pads and the wheel. This friction opposes motion and reduces speed, eventually stopping the bicycle.

5. C) Smooth surfaces have less friction

Smooth floors offer less resistance (friction), so objects move easily and travel farther. Rough carpets have more friction, which slows down motion quickly.

6. A) walking safely

Muscular force pushes your foot against the ground to propel you forward, while **frictional force** provides the necessary grip to keep you from slipping.

7. B) Stretching a rubber band

Applying force stretches the rubber band and changes its shape. This is a clear example of force causing deformation.

8. A) Magnetic force

Magnetic force is a non-contact force that acts between a magnet and magnetic materials like iron, attracting them without physical contact.

9. A) Rough surfaces have more friction

Rough roads provide greater friction, giving better grip to our feet and preventing slipping. Smooth surfaces have less friction and are more slippery.

10. B) A force between charged objects

Electric force acts between charged particles (positive and negative). Like charges repel and unlike charges attract.

Worksheet – 2

1. A) A wheel and axle and a lever

Such devices (like tools with handles and rotating parts) commonly combine a **wheel and axle** (for rotation) and a **lever** (to apply force), making work easier by reducing effort.

2 C) Fulcrum

In a lever, the **fulcrum** is the fixed pivot point about which the lever rotates.

3. B) Axle

Just as a lever rotates around a **fulcrum**, a wheel rotates around an **axle**.

4 A) Inclined plane

An inclined plane (like a ramp) reduces effort by increasing distance, allowing heavy objects to be moved up gradually.

5 B) Scissors
Scissors act as a **double lever**, where two levers are joined at a common fulcrum.

6 A) Pulley
Pulling a rope downward to lift an object upward is the working principle of a **pulley**, which changes the direction of force.

7 A) Turning a door knob
A doorknob works on the **wheel and axle** principle—turning the knob (wheel) rotates the spindle (axle).

8 D) Ramp
Seesaw, bottle opener, and scissors are **levers**, while a ramp is an **inclined plane**, making it the odd one out.

9 B) The object that needs to be moved or lifted
In simple machines, **load** refers to the object or resistance that needs to be moved.

10 C) Wheel and axle
They all feature a larger wheel attached to a the axle that rotate together to transfer force and make movement or turning easier.

Competitive Corner

1 C) Wind force
Magnetic, muscular, and gravitational are standard forces. "Wind force" is not a basic classified type; it is an effect of moving air.

2 B) Wedge
A wedge is used to cut or split objects (e.g., knife, axe).

3 B) Lever
A bottle opener works as a lever, using a fulcrum to lift the cap.

4 B) Pulley
A pulley helps draw water from a well by changing the direction of force.

5 B) Lever
A long tool used under an object to lift it is a lever (works using fulcrum, effort, load).

6 C) To join objects together
A screw fastens objects by converting rotational motion into linear motion.

7 A) Pulley and wheel and axle
Wells commonly use a pulley (rope system) and wheel–axle mechanism to lift buckets.

8 A) Friction
Friction between pencil tip and paper allows writing.

9 C) Class third lever:
because when using a fork, the effort (applied by your fingers) is located between the fulcrum (the base of your hand) and the load (the food on the prongs).

10 A) Both true
Pulley changes direction of force (true). Screw is not a wedge; it is an inclined plane wrapped around a cylinder (false). Hence both evaluations are correct.

ONE OR MORE THAN CORRECT OPTION TYPE

11 A, B, C, D
All options involve applying force—push, pull, magnetic attraction, and writing (muscular force).

12 A, D

Brakes and stopping car involve friction. Options B and C are gravitational and magnetic forces.

13 A, B

Wheelbarrow uses a lever (handles) and wheel & axle (wheel).

14 A, B, C

Pulley, inclined plane, and wedge reduce effort. Chair is not a simple machine.

Assertion & Reason Type

15 A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

Explanation: Friction provides grip between feet and ground, enabling walking.

16 A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

Explanation: Pulley changes direction of force and can reduce effort, making lifting easier.

Matrix Matching Type

17 1–B, 2–C, 3–A, 4–D

Explanation: Effort → applied force; Load → object lifted; Fulcrum → fixed point; Lever → rigid bar.

18 1–B, 2–A, 3–C, 4–D

Explanation: Axe → wedge; Well → pulley; Bottle cap → screw; Ramp → inclined plane.

Comprehension Type

19 B) Lever

Explanation: A seesaw is a lever rotating about a fulcrum.

20 C) Fulcrum

Explanation: The central pivot point of a seesaw is the fulcrum.

21 B) Sit farther from the fulcrum

Explanation: By sitting farther away from the fixed point (fulcrum), Rohan increases the length of his effort arm, making it easier to lift the load on the other side.

22 B) Pulley

Explanation: A pulley is used to lift loads by pulling a rope.

23 B) It changes the direction of the force and makes lifting easier

Explanation: Pulling down lifts the load up, reducing effort.

24 B) Yes, but it would require more effort

Explanation: Without pulley, direct lifting requires greater force.

Previous year questions

1 C. Muscular and elastic force

An archer uses **muscular force** to pull the bowstring and the **elastic force** of the stretched bow helps the arrow move forward towards the target.

2 B. Frictional force

When a ball rolls on the ground, **frictional force** acts opposite to its motion and gradually slows it down until it stops.

3 C) S

if a pushing force is applied at R (from the top right), the ball will move in a straight line away from that push toward S (the bottom left).

4 (B) **repel if same poles face each other**
same side/poles of magnet repel each other

5 **D. See-saw**
A see-saw works on the principle of a **lever**, not a wheel and axle. The other options use wheel and axle mechanisms.

10. Sun, Moon and Earth

Worksheet – 1

1. **Answer: b) Cassiopeia**
   Cassiopeia is a northern constellation whose five brightest stars form a distinct, easily recognizable "W" (or "M" depending on the time of night) shape.
2. **Answer: B) Mars**
   Mars gets its reddish appearance from a high concentration of iron oxide (rust) covering its surface and suspended in its thin atmosphere.
3. **Answer: B) Finding directions**
   A compass uses a magnetized needle that aligns with Earth's magnetic field to always point toward the magnetic North Pole, helping travelers navigate.
4. **Answer: B) Jupiter**
   Jupiter is a massive gas giant with a mass more than two and a half times that of all the other planets in the Solar System combined.
5. **Answer: B) 8**
   There are 8 official planets orbiting the Sun.
6. **Answer: C) Jupiter**
   The Great Red Spot is a colossal, high-pressure counter-clockwise storm in Jupiter's atmosphere that has been observed for hundreds of years.
7. **Answer: D) Mercury**
   Mercury is the innermost planet of our Solar System, orbiting at an average distance of about 58 million kilometers (36 million miles) from the Sun.
8. **Answer: C) The Moon**
   A natural satellite is any celestial body that orbits a planet. Earth has exactly one, which is simply named the Moon.
9. **Answer: C) Uranus**
   Uranus has an extreme axial tilt of 97.8 degrees. This unique orientation causes its poles to face the Sun directly during its orbit, making it look like it's rolling on its side.
10. **Answer: B) Venus**
    Venus is called Earth's twin because it shares a very similar size, mass, density, and basic composition, even though its surface environment is radically different and hostile.

Worksheet – 2

1. **Answer: A) Solar and lunar eclipses**
   The two primary types of eclipses seen from Earth involve the alignment of the Sun, Earth, and Moon. A solar eclipse occurs when the Moon blocks the Sun, and a lunar eclipse occurs when Earth blocks sunlight from hitting the Moon.
2. **Answer: C) When the Earth comes between the Sun and the Moon**
   For a lunar eclipse to happen, the Earth must align directly between the Sun and the Moon, casting its shadow across the Moon's surface.
3. **Answer: B) Moon’s orbit around Earth**
   As the Moon travels around the Earth, the angle at which we see its sunlit half changes. This creates the predictable cycle of lunar phases.

4. Answer: B) Red
During a total lunar eclipse, the Earth's atmosphere filters and bends (refracts) sunlight, scattering blue light away but allowing longer red wavelengths to pass through and light up the Moon. This is often called a "Blood Moon."

5. Answer: B) It reflects sunlight
The Moon is not a star and does not generate its own light. It acts like a giant mirror, reflecting light from the Sun down to Earth.

6. Answer: C) Rock
The Moon is a terrestrial, rocky body. Its surface is covered in a layer of crushed rock and dust called regolith, with a composition similar to Earth's mantle.

7. Answer: B) Summer
Tilting toward the Sun means that a hemisphere receives more direct solar radiation and experiences longer daylight hours, resulting in warmer summer temperatures.

8. Answer: B) 23.5°
Earth does not sit straight up and down relative to its orbital path; it is tilted at an angle of approximately 23.5°, which is the root cause of our seasons.

9. Answer: B) Axis
The Earth rotates once every 24 hours around an imaginary rod running from the North Pole to the South Pole, known as its rotational axis.

10. Answer: B) Waxing Gibbous
"Gibbous" refers to any moon phase where more than half of the lit side is visible. "Waxing" means the lit portion is actively growing larger as it approaches a full moon.

Worksheet – 3

1. Answer: C) Aryabhata
It was India's first indigenously designed and built satellite. It was launched into space on April 19, 1975, using a Soviet Union launch vehicle.

2. Answer: B) Diwali
Diwali, the festival of lights, is traditionally celebrated on Amavasya (the new moon night) of the Hindu lunar month of Kartika, marking the darkest night of that month.

3. Answer: B) Communication
Satellites that relay signals specifically for telephony, television broadcasting, and internet data across long distances are classified as communication satellites.

4. Answer: B) By rockets
To overcome Earth's strong gravitational pull and reach the high speeds required to stay in space, satellites must be launched using powerful rockets.

5. Answer: A) Orbit
An orbit is the specific, gravitationally curved path that a celestial object or spacecraft continuously follows as it travels around a planet or star.

6. Answer: B) Capricorn
In Vedic astrology, the word "Makar" translates to the zodiac sign Capricorn. The festival marks the exact day the Sun transitions into this constellation.

7. Answer: B) Full Moon
Holi, the festival of colors, is celebrated on Purnima (the full moon day) during the Hindu lunar month of Phalguna.

8. Answer: B) A man-made object sent into space
While celestial bodies like the Moon are natural satellites, any machinery intentionally constructed by humans and launched into space to orbit an object is an artificial satellite.

9. Answer: B) Bhaskara-I
Launched by ISRO in 1979, Bhaskara-I was India's first low-orbit Earth observation satellite. It carried specialized sensors used for hydrology, forestry, and oceanography.

10. Answer: B) 29.5 days
This duration is known as a synodic month. It is the time it takes for the Moon to cycle through all its shapes as seen from Earth, measured from one new moon to the very next.

Competitive Corner

Single correct option Type

1. Answer: B) North–South direction
A compass needle is a small magnet that is free to pivot horizontally. It aligns itself with the Earth's natural magnetic field, pointing towards the magnetic North and South poles.

2. Answer: C) Hot gases like hydrogen and helium
Stars are massive spheres of plasma held together by gravity. They are predominantly composed of hydrogen and helium, which undergo nuclear fusion in the star's core to generate light and heat.

3. Answer: B) 23.5°
The Earth does not spin perfectly upright relative to its orbital plane around the Sun; its axis of rotation is tilted at an angle of approximately 23.5°.

4. Answer: B) summer
When a hemisphere tilts toward the Sun, it receives more direct solar radiation and experiences longer daylight hours, resulting in warmer temperatures and the summer season.

5. Answer: B) both hemispheres receive nearly equal sunlight
These seasons begin during the equinoxes, when the Earth's axis is tilted neither toward nor away from the Sun. As a result, both the Northern and Southern hemispheres receive nearly equal amounts of sunlight.

6. Answer: B) crescent moon
As the Moon moves away from the New Moon phase and begins to reflect light back to Earth, the first sliver of the illuminated surface we see is a thin, curved shape known as a crescent moon.

7. Answer: B) When one celestial object blocks another
An eclipse occurs when one astronomical body moves into the shadow of another or physically blocks light from a star, such as during a solar or lunar eclipse.

8. Answer: A) Venus
Although Mercury is closest to the Sun, Venus is the hottest planet because it possesses a thick, dense atmosphere rich in carbon dioxide, which traps heat through an intense greenhouse effect.

9. Answer: C) North Pole
The Pole Star (Polaris) is located nearly directly above Earth's geographic North Pole. Because it is aligned with Earth's axis of rotation, it appears fixed in position while the rest of the night sky rotates around it.

10. Answer: B) Mars and Jupiter
The vast majority of asteroids in our solar system reside within the main Asteroid Belt, a vast ring-shaped region situated between the orbits of Mars and Jupiter.

ONE OR MORE THAN CORRECT OPTION TYPE

11. **Answer:** A and C
A is correct because the Moon is Earth's only natural satellite and orbits it.
C is correct because the Moon has a negligible exosphere, which is practically considered a vacuum (no atmosphere).

12. Answer: A, C, and D
A is correct because the Sun is a main-sequence star.
C is correct because it accounts for about 99.8% of the total mass of the Solar System.
D is correct because its immense gravitational pull keeps all the planets orbiting around it.

13. Answer: C
C is correct because Saturn (along with Jupiter) is classified as a gas giant, primarily composed of hydrogen and helium.

14. **Answer:** A, B, and D
A is correct (Mars has 2 small moons: Phobos and Deimos).
B is correct (Earth has 1 moon).
D is correct (Jupiter has dozens of confirmed moons, including Europa, Ganymede, and Callisto).

Assertion and Reason Type

15. Answer: C) Assertion (A) is true and Reason (R) is false
Assertion (A) is True: The Sun is indeed at the center of our solar system.
Reason (R) is False: The Sun's gravity absolutely *does* pull all the planets toward it; this immense gravitational pull is exactly what keeps the planets orbiting around it instead of floating off into space.

16. Answer: C) Assertion (A) is true and Reason (R) is false
Assertion (A) is True: The tilt of the Earth's axis is the primary reason we experience different seasons as the planet orbits the Sun.
Reason (R) is False: The Earth's axis does not remain upright; it is tilted at an angle of approximately 23.5 degree

MATRIX MATCHING TYPE

17. Correct Matches:
A → R, B → P, C → Q, D → S

- Holi is celebrated on the full moon day (Purnima) of the Hindu month of Phalguna.
- Lunar New Year begins with the first new moon of the lunisolar calendar.
- Easter is celebrated on the first Sunday after the first full moon following the vernal equinox.
- Harvest Festivals are often linked to specific full moons or star alignments that indicate crop readiness.

18. Correct Matches:
A → R, B → P, C → S, D → Q

- Jupiter is the largest planet in the Solar System.
- Saturn is famous for its spectacular ring system.
- Uranus rotates on its side, making it unique among the planets.

- The Water Cycle is the continuous movement of water between Earth's surface, atmosphere, and underground reservoirs.

COMPREHENSION TYPE

Passage 1

19. Answer: B) It causes tides due to its gravitational pull

The Moon's gravitational force pulls on Earth's oceans, causing the regular rise and fall of sea levels known as tides.

20. Answer: C) Meteorite impacts

Most lunar craters were formed when meteorites and asteroids struck the Moon's surface over millions of years.

21. Answer: C) 29.5 days

The Moon takes approximately 29.5 days to complete one orbit around Earth, which corresponds to the lunar month.

Passage 2

22. Answer: B) Light and heat

The Sun supplies light and heat, which are essential for life processes such as photosynthesis and maintaining suitable temperatures on Earth.

23. Answer: C) Gas

The Sun is a giant ball of hot gases, primarily hydrogen and helium.

24. Answer: B) Nuclear fusion

Inside the Sun's core, hydrogen atoms combine to form helium through nuclear fusion, releasing enormous amounts of energy in the process.

Previous year questions

1. Answer: D) Both A and B

Eid-ul-Fitr is celebrated upon the sighting of the Crescent Moon.
Diwali is celebrated on the night of Amavasya, which is the New Moon night.
Since both A and B are correct matches, option D is the correct choice.

2. Answer: B) 23.5°

The Earth does not orbit completely upright; its rotational axis is tilted at an angle of approximately 23.5° relative to its orbital plane around the Sun. This tilt is responsible for our changing seasons.

3. Answer: C) c

In the diagram, label 'a' points to the Sun, label 'b' points to the Earth, and label 'c' points directly to the Moon as it passes through the Earth's shadow.

4. Answer: C) Moon, Earth, Sun

"Ascending order" means arranging from the smallest to the largest size. The Moon is smaller than the Earth, and the Sun is by far the largest object among the three.

5. Answer: C) Earth's magnetic field

A compass contains a tiny, lightweight magnet (the needle) that is free to pivot. This needle aligns itself naturally with the Earth's magnetic field, pointing towards the magnetic North Pole.

11. Air and Water

Worksheet - 1

1. Correct Answer: D
Air contains about 21% oxygen. The rest is mostly nitrogen (~78%) with small amounts of other gases.

2. Correct Answer: A
Nitrogen makes up the largest portion of air (about 78%), so it is the major component of the atmosphere.

3. Correct Answer: A
The troposphere is the lowest layer, and the stratosphere has the ozone layer. The exosphere is actually the outermost layer, not the lowest.

4. Correct Answer: B
Oxygen supports combustion, which means it helps substances burn.

5. Correct Answer: C
Humans and animals breathe in oxygen because it is essential for respiration and releasing energy from food.

6. Correct Answer: C
Oxygen both supports burning and is used by living organisms for respiration. Hence, it fits in the overlapping region (X) of the Venn diagram.

7. Correct Answer: B
The envelope of air surrounding the Earth is called the atmosphere. It contains gases like nitrogen, oxygen, and carbon dioxide.

8. Correct Answer: B
A candle needs oxygen to burn. Inside the closed glass, oxygen gets consumed, and once it is insufficient, the flame goes out.

9. Correct Answer: B
Air contains about 78% nitrogen and 21% oxygen. This is approximately a 4:1 ratio of nitrogen to oxygen.

10. Correct Answer: C
The tiny shining particles seen in a beam of sunlight are dust particles. They scatter light, making them visible to our eyes.

Worksheet – 2

1. Correct Answer: C
Water is made of two hydrogen atoms and one oxygen atom, so its chemical formula is H_2O.

2. Correct Answer: A
Water is essential for seed germination and is widely used in industries. It is not used as a fuel for burning.

3. Correct Answer: B
Water freezes at 0°C under normal conditions, turning from liquid into solid ice.

4. Correct Answer: B
In sedimentation, heavier particles like mud settle at the bottom when water is left undisturbed.

5. Correct Answer: C
Water does not burn; instead, it is often used to put out fires.

6. **Correct Answer: B**
Condensation occurs when water vapour cools and changes into tiny droplets, forming clouds.

7. **Correct Answer: C**
Oceans and seas contain most of Earth's water, but it is salty and not suitable for drinking.

8. **Correct Answer: C**
Evaporation is the process where water changes from liquid to vapour when heated.

9. **Correct Answer: B**
Groundwater forms when rainwater seeps into the soil and collects in underground layers.

10. **Correct Answer: B**
First, the mud settles at the bottom when the water is left undisturbed (sedimentation). Then, the clear water is carefully poured off from the top (decantation).

Worksheet – 3

1. **Correct Answer: B**
Salt dissolves completely in water, making it a dissolved impurity. Sand, mud, and pebbles are insoluble impurities.

2. **Correct Answer: A**
Filtration removes solid (insoluble) impurities like sand and dirt from water using a filter.

3. **Correct Answer: B**
Bacteria are living (biological) impurities that can contaminate water and cause diseases.

4. **Correct Answer: B**
Evaporation is the process where water changes into vapour due to heat from the Sun.

5. **Correct Answer: B**
First, water vapour cools to form clouds (condensation), then the Sun heats water (evaporation), followed by precipitation, and finally collection/flow back to oceans.

6. **Correct Answer: C**
Reusing water helps reduce wastage and conserves this valuable resource. The other options promote water wastage.

7. **Correct Answer: B**
Rainwater harvesting collects and stores rainwater for later use, helping in water conservation.

8. **Correct Answer: A**
Water is essential for digestion and other body functions. It does not directly give energy, increase weight, or stop breathing.

9. **Correct Answer: B**
Wet clothes dry faster outside because heat from the Sun increases evaporation of water.

10. **Correct Answer: B**
Using water wisely helps conserve water and prevents wastage. The other options lead to water pollution or unnecessary loss of water.

Competitive Corner

Single Option Correct Type

1. Correct Answer: C
The atmosphere is the layer of air that surrounds the Earth. It contains gases like nitrogen and oxygen that support life.

2. Correct Answer: B
The troposphere is the lowest layer of the atmosphere and is closest to Earth's surface. Weather changes occur in this layer.

3. Correct Answer: B
Oxygen is essential for breathing and is used by living organisms during respiration to release energy.

4. Correct Answer: B
Nitrogen makes up about 78% of the air, making it the major component of the atmosphere.

5. Correct Answer: B
Windmills work because moving air (wind) has energy that can turn the blades and help draw water.

6. Correct Answer: C
Plants use carbon dioxide during photosynthesis to make food in the presence of sunlight.

7. Correct Answer: B
Carbon dioxide does not support combustion and is used in fire extinguishers to put out fires.

8. Correct Answer: C
Water is used for cooking, washing, and irrigation, but it does not help in burning; instead, it extinguishes fire.

9. Correct Answer: B
Water vapour is the gaseous state of water, and ice forms at 0°C. However, water also exists in the gaseous state, so statement (iii) is false.

10. Correct Answer: C
Chlorine is added to drinking water to kill harmful bacteria and viruses, making it safe to drink.

11. Correct Answer: C
Rainwater harvesting helps collect and store rainwater, which replenishes groundwater and prevents wastage.

12. Correct Answer: C
The troposphere is the lowest layer of the atmosphere, extending up to about 8–15 km (around 11 km on average). It is the layer where most weather phenomena like clouds, rain, and storms occur.

One or More than One Correct Option Type:

13. Correct Answer: A, B, C, D
Air is a mixture of gases. It mainly contains nitrogen and oxygen, with small amounts of carbon dioxide and traces of other gases like hydrogen.

14. Correct Answer: A, B, C
The atmosphere protects Earth from harmful radiation, provides oxygen for breathing, and helps maintain temperature. It does not increase pollution; pollution is caused by human activities.

15. Correct Answer: A, B, C
Water is used for drinking, cooking, and generating electricity (hydropower). It is not used for burning; instead, it helps extinguish fire.

16. Correct Answer: A, B, C, D
Water can be purified by boiling (kills germs), sedimentation (settles impurities), filtration (removes particles), and evaporation (separates pure water as vapor).

17. Correct Answer: A, B, C
The water cycle includes evaporation, condensation, and precipitation. Combustion is not part of the water cycle.

18. Correct Answer: A, B, D
Water can be conserved by rainwater harvesting, fixing leaks, and reusing water. Using excess water leads to wastage, not conservation.

Assertion and Reason Type

19. Correct Answer: A
The atmosphere protects us from harmful solar radiation, and this is mainly due to the ozone layer absorbing harmful ultraviolet (UV) rays. Therefore, both statements are true and the reason correctly explains the assertion.

20. Correct Answer: A
Rainwater harvesting is an effective method of water conservation. It helps recharge groundwater and reduces wastage, so the reason correctly explains the assertion.

Matrix Matching Type

21. Correct Answer: a-S, b-P, c-Q, d-R
Boiling involves heating water to kill harmful bacteria. Windmills are used to generate electricity. Air is a mixture of gases. Chlorination means adding chemicals to kill germs.

22. Correct Answer: a-R, b-P, c-S, d-Q
Troposphere is the lowest layer where weather changes occur. Stratosphere contains the ozone layer that absorbs UV rays. Thermosphere reflects radio waves in ionosphere region. Exosphere is the uppermost layer merging into space.

Comprehension Type

Passage – I

23. Correct Answer: B
The passage clearly states that the atmosphere is a layer of gases that surrounds the Earth.

24. Correct Answer: C
Nitrogen makes up about 78% of the atmosphere, which is the largest proportion among all gases present.

25. Correct Answer: B
The atmosphere protects living beings from harmful UV rays and supplies oxygen needed for respiration, making life possible on Earth.

Passage – II

26. Correct Answer: B
The passage states that water is becoming scarce because of increasing population and pollution.

27. Correct Answer: B
The passage mentions that people waste water in daily activities like washing and bathing.

28. Correct Answer: B

Rainwater harvesting is mentioned as an effective method to conserve water.

Previous Year Questions

1. Correct Answer: A

The presence of the Moon indicates nighttime conditions. At night, land cools faster than water, so the air over land becomes cooler and denser. This causes wind to blow from land (P) to water (Q), which is called a land breeze.

2. Correct Answer: A

Clothes dry due to evaporation, which is labeled as P in the water cycle. Floods are caused by precipitation (heavy rainfall), which is labeled as R.

3. Correct Answer: B

After water reaches its boiling point (100°C), continued heating does not increase the temperature. Instead, water starts converting into steam, so the volume of liquid water decreases. Hence, the correct figure shows 100°C with reduced volume (65 mL).

4. Correct Answer: A

The balloon filled with air has mass and is heavier than the empty balloon. Therefore, the side with the inflated balloon goes down, and the empty balloon side goes up, as shown in option (A).

5. Correct Answer: C

In the activity, when the burning candle is covered with a glass jar, the flame eventually goes out. This happens because the oxygen inside the jar gets used up. Without oxygen, combustion cannot continue, showing that air is necessary for burning.

12. Pollution and Natural Disasters

Worksheet – 1

1. **Correct Answer: B**
   Pollution means the presence of harmful substances in the environment that can damage living beings and natural resources.
2. **Correct Answer: C**
   Smoke from factories and cars releases harmful gases into the air, making it polluted.
3. **Correct Answer: B**
   Water pollution occurs when harmful wastes like industrial discharge contaminate rivers, lakes, and oceans.
4. **Correct Answer: B**
   When nitrogen and sulphur oxides mix with water vapour, they form acids that fall as acid rain.
5. **Correct Answer: C**
   Dumping plastic and waste on land contaminates the soil and harms organisms living in it.
6. **Correct Answer: B**
   Walking or cycling reduces the use of vehicles, thereby decreasing air pollution.
7. **Correct Answer: C**
   Burning fossil fuels in vehicles releases harmful gases, causing air pollution.
8. **Correct Answer: A**
   Statements (i), (ii), and (iv) are correct; pesticides harm soil quality, so (iii) is incorrect.
9. **Correct Answer: C**
   Reducing plastic use and recycling helps keep soil clean and prevents pollution.
10. **Correct Answer: B**
    Hot water reduces the amount of dissolved oxygen in water, which can kill fish and other aquatic life.

Worksheet – 2

1. **Correct Answer: B**
   A flood is a natural disaster caused by excessive water overflowing onto land.
2. **Correct Answer: B**
   Earthquakes are caused by the sudden movement of tectonic plates beneath the Earth's surface.
3. **Correct Answer: B**
   Taking shelter in a safe place helps protect people from strong winds and heavy rain during a hurricane.
4. **Correct Answer: B**
   Statements (i), (ii), and (iii) are correct; volcanic eruptions *can* cause natural disasters, so (iv) is incorrect.
5. **Correct Answer: C**
   Cutting down trees harms the environment, while the other options help protect it.
6. **Correct Answer: C**
   Using energy-efficient bulbs reduces electricity consumption and saves energy.
7. **Correct Answer: B**
   Trees improve air quality by absorbing carbon dioxide and releasing oxygen.

8. **Correct Answer: C**
Hot, dry conditions can lead to forest fires, which are called wildfires.

9. **Correct Answer: C**
Deforestation mainly happens due to cutting trees for wood and clearing land for farming.

10. **Correct Answer: D**
National Cadet Corps (NCC) is an organization, while the others are natural disasters.

Competitive Corner

Single Option Correct Type

1. **Correct Answer: B**
Throwing plastic and waste on the ground contaminates soil and harms living organisms.

2. **Correct Answer: B**
Using public transport reduces the number of vehicles, lowering air pollution.

3. **Correct Answer: A**
A flood occurs when water levels rise suddenly and overflow onto land.

4. **Correct Answer: A**
Planting trees helps improve air quality and supports the environment.

5. **Correct Answer: B**
Natural disasters often damage homes, roads, and other infrastructure.

6. **Correct Answer: A**
Earthquakes are caused by the sudden release of energy beneath the Earth's surface.

7. **Correct Answer: B**
Water, food, and first-aid supplies are essential for survival during emergencies.

8. **Correct Answer: B**
Chemicals and waste discharged into water bodies pollute water and harm aquatic life.

9. **Correct Answer: C**
Air pollution can cause breathing problems such as asthma and other lung diseases.

10. **Correct Answer: B**
Destroying mangroves and coral reefs disrupts the balance of ecosystems.

11. **Correct Answer: D**
Planting trees does not pollute water; it actually helps protect the environment.

One or More than One Correct Option Type:

12. **Correct Answer: A, B, C**
Vehicles release harmful gases, industrial waste pollutes water, and excessive pesticides contaminate soil. Planting trees helps reduce pollution.

13. **Correct Answer: A, B, C**
Recycling, saving electricity, and proper waste disposal protect the environment. Increasing CFCs harms the ozone layer.

14. **Correct Answer: A, B, C**
Natural disasters can destroy infrastructure, cause loss of life, and damage wildlife. They do not improve ecosystems immediately.

15. Correct Answer: A, B, D
Smog is a mix of smoke and fog, makes breathing difficult, and is a form of air pollution. It does not form only in oceans.

16. Correct Answer: B
Wildfires can start due to lightning or extreme heat and spread quickly in dry conditions.

17. Correct Answer: B
Trees and vegetation hold soil together and absorb harmful substances, preventing soil erosion.

18. Correct Answer: A
Coral reefs are aquatic ecosystems that are highly affected by water pollution.

19. Correct Answer: B
A tornado is a fast-moving rotating column of air that causes destruction.

Assertion and Reason Type

20. Correct Answer: B
Both statements are true, but the reason is not the correct explanation. Breathing problems are mainly caused by harmful pollutants like carbon monoxide, sulphur dioxide, and particulate matter—not carbon dioxide.

21. Correct Answer: C
The assertion is true because natural disasters can damage ecosystems for a long time. The reason is false since disasters do not always improve the environment; they often cause harm.

Matrix Matching Type

22. Correct Answer: a-P, b-R, c-Q, d-S
Natural disasters damage lives and property (P). Pollution harms the environment (R). Environmental protection helps conserve resources (Q). Deforestation leads to loss of biodiversity (S).

23. Correct Answer: a-S, b-R, c-P, d-Q
Air pollution involves smoke and harmful gases (S). Water pollution is contamination of water bodies (R). Soil pollution includes harmful chemicals in soil (P). Deforestation means cutting down trees (Q).

Comprehension Type

Passage – I

24. Correct Answer: C
Earthquakes are mainly caused by the movement of tectonic plates beneath the Earth's surface.

25. Correct Answer: C
Hurricanes are powerful storms that often affect coastal areas with strong winds and heavy rainfall.

26. Correct Answer: B
Floods occur when rivers overflow due to heavy rainfall, covering surrounding land with water.

Passage – II

27. Correct Answer: C
Burning fossil fuels releases harmful gases and particles into the air, causing air pollution.

28. Correct Answer: B
Water pollution happens when harmful chemicals and waste enter rivers, lakes, and oceans.

29. Correct Answer: B
Dumping plastic and waste on land contaminates soil and harms plants and animals living in it.

Previous Year Questions

1. Correct Answer: C
Oxides of nitrogen and sulphur react with water vapour to form acids, which fall as acid rain.

2. Correct Answer: D
All listed organisations (NCC, WHO, and Red Cross) contribute to disaster management and relief activities.

3. Correct Answer: B
(ii) is incorrect because terrace farming is done in hilly areas, not deserts. (iv) is incorrect because such storms in the USA are called hurricanes, not typhoons.

4. Correct Answer: B
Planting trees helps the environment, while the other activities cause harm.

5. Correct Answer: D
Both staying under a table indoors and staying away from structures outdoors are correct safety measures during an earthquake.